

BLAST Sequence Analysis Report

Analyst: Bharat

Date: June 2026

Institution: BSc Biotechnology, KK Shastri College

SEQUENCE ANALYZED

Gene: BRCA1 (Breast Cancer Susceptibility Gene 1)

Organism: Homo sapiens

Source: NCBI Accession NM_007294

BLAST PARAMETERS

Tool: Nucleotide BLAST (blastn)

Database: nr (non-redundant)

Website: blast.ncbi.nlm.nih.gov

RESULTS

Top Match: Homo sapiens BRCA1 DNA repair associated mRNA

% Identity: 100%

E-value: 0.0

Query Coverage: 100%

INTERPRETATION

% Identity of 100% confirms exact match with known human BRCA1 sequence. E-value of 0.0 indicates extremely high confidence - zero probability this match occurred by random chance.

CONCLUSION

Sequence conclusively identified as human BRCA1 gene, associated with DNA repair and hereditary breast and ovarian cancer susceptibility.

Prepared by Bharat

github.com/bharat-bioinfo